CLASSIFICATIONS AND MICROORGANISMS

CONTENT

- Biology
- 5 Kingdom Classifications By R.H. Whittaker
- Binomial Nomenclature
- Monera
- Bacteria

BIOLOGY

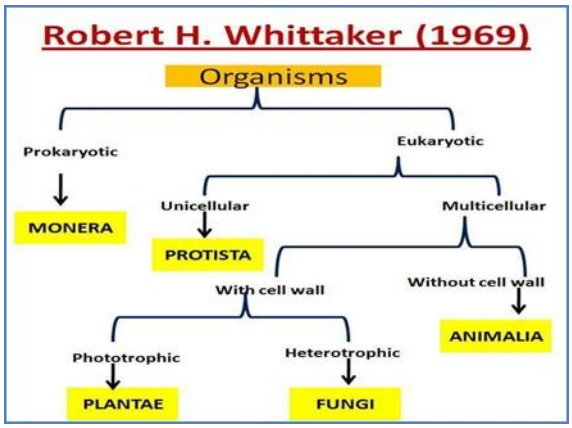
- The study of living beings is called biology.
- Father of biology – Aristotle
- Father of botany – Theophrashtus
- Father of zoology – Aristotle
- What is living organisms?
  - **Growth** – The increase in number of cells or mass is called growth.
  - **Reproduction**- Living organisms produce young ones of the same kind
  - **Metabolism** – Chemical reaction occurring inside a living cell.
  - **Development** – Maturity in the body

TYPES OF METABOLISM REACTIONS

- **Anabolism** – The process of formation of larger molecules from smaller molecules are called anabolism.
  - Example: Photosynthesis 4r
- **Catabolism** – The process of breaking down of larger molecules into smaller with the release of energy.
  - Example: Respiration
- In **Linnaeus'** time a Two Kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively

Book: Systema Nature LIMITATIONS

- **Limitations** of two kingdom classification:
  - Not covered Eukaryotes and Prokaryotes.
  - Not covered unicellular and multicellular organisms.
  - Not covered photosynthetic and non photosynthetic.
- A new classification came into existence by **R.H. whittaker** based on cell structure, thallus structure, thallus organization, mode of nutrition, reproduction and **phylogenetic relationship** (based on evolutionary relationship).
- After 5 kingdom classifications there is another type of classification was given by **Carl Woese** who gave **6 kingdom classification**. He separated prokaryotes into Eubacteria and Archaebacteria.



5 KINGDOM CLASSIFICATIONS BY R.H. WHITTAKER

MONERA

- Prokaryotic, unicellular without nucleus.
  - Example: Bacteria, cynobacteria: Blue Green Algae.

PROTISTA

- Eukaryotic, unicellular with nucleus.
- Mainly found in aquatic.
  - Example: Diatoms, Algae and Protozoa.

FUNGI

- Multicellular, found on dead and decaying matter.
  - Example: Mushroom : edible
- **Taxonomy classification : (Father of Taxonomy is Linneaus)**
- **Trick: Khoob Padenge Class OF GS**
- **Kingdom---Phylum---Class---Order---Family---Genus---Species**

	Monera	Protista	Fungi	Plantae	Animalia
Type	Unicellular Prokaryotes	Unicellular Eukaryotes	Multicellular Non green Eukaryotic	Multicellular, Eukaryotic	Multicellular Eukaryotic
examples	Bacteria, Blue-green Algae 	Amoeba, Paramecium, Euglena 	Yeast, Rhizopus, Mushrooms moulds 	Trees, Plants, Shrubs 	Fish, Insects, Animals like elephant, Humans, Birds 

BINOMIAL NOMENCLATURE

- In 1730s – Swedish botanist – **Carlous Linnaeus** – developed two word naming system.
- In binomial nomenclature, each species is assigned a 2 part scientific name, written in italic.
- The First word begins with capital letter = Genus (group of similar species) to which organism belongs.
- The Second word is lowercased = unique to each species.
- Genus comprises a group of related species which has more characters in common in comparison to species of other genera.
- We can say that genera are aggregates of closely related species. For example, Lion (*Panthera leo*), leopard (*pardus*) and tiger (*P. tigris*) with several common features, are all species of the genus Panthera
- A species is a group of individuals capable of interbreeding and producing fertile offspring.
- Biological names are generally in Latin and written in **italics**. They are Latinised or derived from Latin irrespective of their origin.
- The first word in a biological name represents the genus while the second component denotes the specific epithet.
- Book: species planatarum

DESCRIPTION OF IMPORTANT OR HABITAT

- Banyan – Ficus benghalensis
- Pea plant – Pisum sativum
- Human beings – Homo sapiens
- Mango- Magnifiera indica
- Onion- Allium cepa
- Potato – Solanum tubersum

COMMON SCIENTIFIC NAMES

- Cat – Felis catus
- Dog – Cannis familiaris
- Frog – Rana hexadactyla/Rana tigrina

- Housefly – Musca domestica
- Horse – Eqqus caballus
- Lion – Panthera leo
- Tiger – Panthera tigris
- Honey bee – Apis indica
- Rose-- Rosa Centifolia
- Cockroach—Periplanta Americana
- Buffalo--- Bubalis babalis
- Cow- Bos Indicus
- Human- Homo Sapiens
- Wheat—Triticum Aestivum
- Rice—Oryza Sativum
- Neem- Azadirachta Indica
- Imli—Tamarindus Indica
- Turmeric-- Curcuma longa

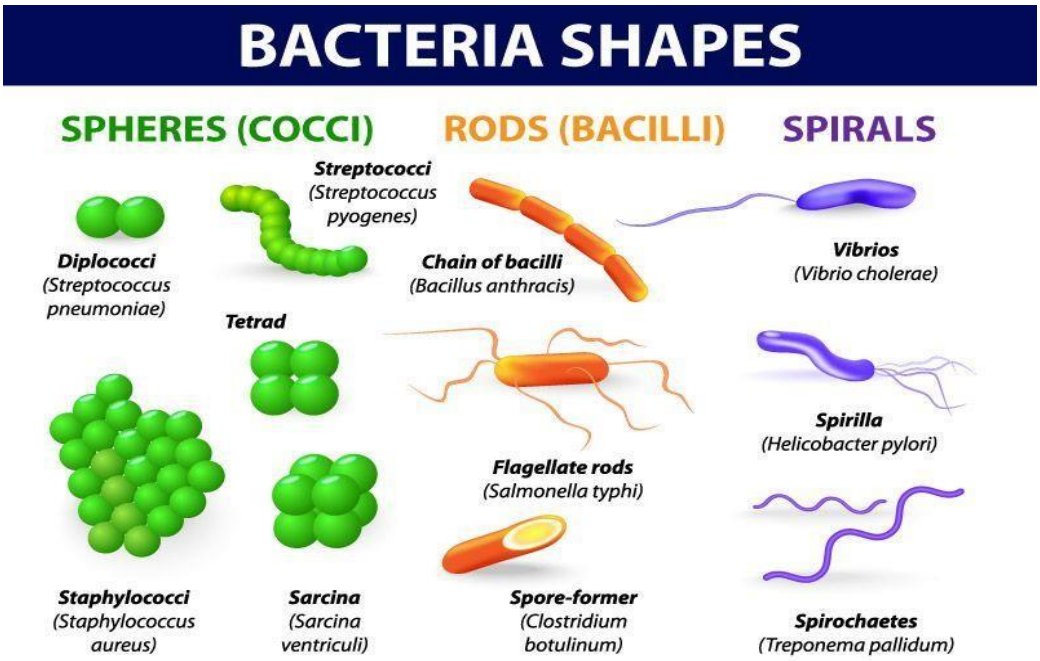
MONERA

- The organisms in this group are
- **Prokaryotes** do not have a defined nucleus or organelles {Prokaryotic Cells vs. Eukaryotic Cells}.
- **Unicellular** do any of them show multi-cellular body designs.
- This group includes all Some well-known bacteria include **blue-green algae** or **cyanobacteria** [have cell walls], and **mycoplasma** [doesn't possess a Cell Wall].
- They are the most abundant micro-organisms and live in extreme habitats.
- The mode of nutrition of organisms in this group can be either by synthesizing their own food (**autotrophic**) or getting it from the environment (**heterotrophic**).
- Many of them live in or on other organisms as **parasites**.

BACTERIA

- Father of Bacteriology is Louis Pasteur.
- Ehrenberg used the word bacteria.
- Bacteria are microscopic, single-celled organisms that thrive in diverse environments.
- These organisms can live in soil, the ocean and inside the human gut. Humans' relationship with bacteria is complex.
- Sometimes bacteria lend us a helping hand, such as by curdling milk into yogurt or helping with our digestion.
- Bacteria are prokaryotes. Hence they do not have a well defined nucleus and have a cell wall composed of peptidoglycan, a protein sugar molecule.
- Their genetic material, which consists of circular and double-stranded DNA, floats freely in the cytoplasm and is not membranebound.
- The region in the cytoplasm, where the genetic material floats is known as nucleoid.
- Louis Pasteur discovered the Rabies (hydrophobia) bacteria. He also discovered the process of Pasteurization of milk.
- Low temp. 62.8 degree Celsius for 30 min
- High temp 71.7 degree Celsius for 15 seconds.
- Robert Koch discovered tuberculosis bacteria.
- Vaccine is BCG (Bacillus Calmette Guerin)
- The common structural forms are:-
    - Spherical or ovoid (Coccus) Ex: Streptococcus thermophillus
    - Rod shaped or cylindrical (Bacillus) Ex: Bacillus bulgaricus

- Spiral or screw (Spirillum) Ex: Borrelia burgdorferi
- Comma shaped (Vibrio) Ex: Vibrio Cholera



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MICROORGANISM

CONTENT

- Bacteria Used In Nitrogen Fixation
- Bacteria In Nitrification
- Oil Zapper
- Cyanobacteria
- Archaebacteria (Intermediate Between Bacteria And Eukaryotes)
- Eubacteria
- Kingdom Protista
- Types Of Algae
- Economic Importance Of Algae

BACTERIA USED IN NITROGEN FIXATION

- Free living aerobic bacteria: Azotobacter
- Free living anaerobic bacteria: Clostridium– living free in the soil and fix nitrogen,purple sulphur bacteria
- Symbiotic in legumes and pulses: Rhizobium (found in root nodules)
    - Examples: Beans
- Symbiotic in sugarcane: Glucoacetobacter diazotrophicus (found in stem knots)
- Symbiotic in other plants: Frankia, Azospirillum
- E.coli (Escherichia coli) bacteria in large intestine helps in the formation of Vit K andvitamin B complex
- Similarly, Clostridium butyclicum is used for commercial preparation of riboflavin, andvitamin B.

BACTERIA IN NITRIFICATION

- Nitrification is the process in which the ammonia is converted into Nitrate. Nitrificationis a two step process and based upon these two steps, the bacteria are divided into Nitrosifying and Nitrite-Oxidizing bacteria.
- Conversion atmospheric Nitrogen into Nitrite with help of **nitrosomonas** nitrite to nitrate with help of **nitrobacter** bacteria & again nitrate into atmospheric nitrogen withhelp of denitrifying bacteria **pseudomonas**.
    - Example of **Nitrosifying bacteria is Nitrosomonas**, which converts the Ammonia
    - (NH3) into Nitrite (NO2-).
- Example of **Nitrite-Oxidizing bacteria is Nitrobacter** which are able to oxidize theNitrite and crate Nitrate (NO3-).
- **Clostridium acetobutylicum** is able to produce acetone from acetic acid as well asbutanol from butyric acid.
- **Bacillus thuringiensis (also called BT),** a Gram-positive, soil dwelling bacterium is usedfor Pest Control and producing Bt crops.
- Bioremediation techniques such as Oil zapper use bacteria.
- Many **antibiotics** are used from bacteria. Some of them are Bacitracin, Polymyxin B,Streptomycine, Erythromycine, neomycin-B, Chloramophenicol etc.
- **Pleasant smell of the earth** after the first shower (earthy odour) is caused by theproduction of a series of **streptomycete metabolites called geosmins.**

OIL ZAPPER

- 'Oilzapper' technology was developed by ONGC-Teri Biotech Ltd (OTBL), a joint venturebetween ONGC and TERI.
- This technology was first used by OTBL in Mehsana in Gujarat to eliminate an oil spilland manage

the sludge created from the first oil well in the region.
- The water became clean and subsequently a home to a variety of birds.

CYANOBACTERIA

- They are photosynthetic; they can make their own food.
- They are **photosynthetic prokaryotes** and able to produce oxygen.
- They are also called **Blue Green Algae**.
- Found in **aquatic and terrestrial habitat**.
- **Cyanobacteria such as Anabena** (a symbiont of the aquatic fern azolla) can **provide riceplantation with biofertilizer.**

ARCHAEBACTERIA (INTERMEDIATE BETWEEN BACTERIA AND EUKARYOTES)

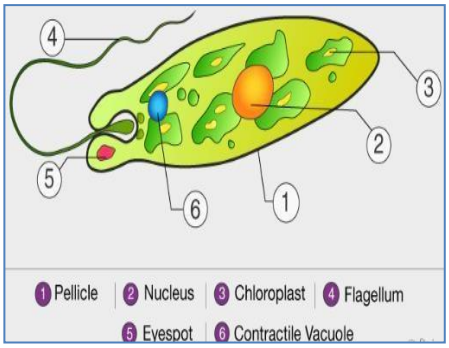
- Lactobacillus helps in formation of milk. (lactose to lactic acid)
- Mycococcus in Tobacco plants
- Bacillus Megatherium present in Tea plants for fragnance.
- **BT (Bacillus Thurengesis) Cotton**: Used For Pest Control in Bio Cotton.
- Bioremediation bacterias performs biological activity, helps in recycling i.e. oil zapper,used in antibiotics.
- **Streptomycete** bacteria also known as geosmin is responsible for the smell in first rainshower.
- **Cynobacteria** i.e. blue green algae they are phoyosynthetic prokaryotes,their wall ismade of pactedoglycon ,both aquatic and terrestrial habitat.
    - Example: **Anabena** has symbiotic association with **Azolla** (aquatic fern), works asbiofertilizer in rice cultivation.
- They play intermediate role between bacteria & eukaryotes.
- Most of them are autotrophs and only a few photosynthesize.
- They derive their energy for metabolic activities, from the oxidation of chemical energysources like reduced gases – Ammonia, methane or hydrogen sulphide.
- In the presence of these chemicals it can manufacture their own amino acids acidproteins.
- These bacteria are special since they live in some of the **most harsh** habitats such asextreme salty areas (**halophiles**), hot springs (**thermoacidophiles**) and marshy areas(**methanogens**) .
- Archaebacteria differ from other bacteria in having a **different cell wall structure** andthis feature is responsible for their survival in extreme conditions.
- **Methanogens** are present in **the gut of several ruminant animals** such as cows and buffaloes and they are responsible for the production of **methane (biogas)** from thedung of these animals.

EUBACTERIA

- There are thousands of different **eubacteria or 'true bacteria'**.
- They are characterized by the presence of a **rigid cell wall**, and if motile, a **flagellum**.

KINGDOM PROTISTA

- This group includes many kinds of unicellular eukaryotic organisms such as unicellular algae, protozoans and **unicellular** fungi.
- Some of these organisms use appendages i.e. hair like structure cilia like in parameciumor whip-like flagellum
    - Example: euglena.
- Their mode of nutrition can be **autotrophic** (Unicellular algae, diatoms) or **heterotrophic**/protozoans. .
    - Examples: Unicellular algae, chlorella, Euglena, Trypanosoma (causes sleeping sickness), Amoeba, Plasmodium, Paramecium,Chlamydomonas etc.



- Brown algae – The Phaeophyta
- Red algae – The Rhodophyta
- Green algae – The Chlorophyta
- Their mode of nutrition can be autotrophic or heterotrophic.
- Members of Protista are **primarilyaquatic**.
- Some have flagella or cilia that helpin movement.
- Protists,reproduce **asexually** and **sexually** by a process involving cell fusion and zygote formation.

TYPES OF ALGAE

- Study of algae is called phycology.
- Most primitive algae are blue green algae.
- Algae have been divided on the **basis of nature of pigments** present in them and the **mode of storing food.** These pigments give them specific color.

GREEN ALGAE OR CHLOROPHYCEAE

- Green algae have **chloroplast and chlorophyll** in their cells.
  - Examples are Chlamydomonas, Volvox, Spirogyra, Ulothrox, Oedogonium andChara are some example.

BROWN ALGAE OR PHAEOPHYCEAE

- Brown algae store food in the form of **laminarin and mannitol.**
- Many of brown algae are called Kelps/sea weeds.
- Ectocarpus, Laminaria, Sargassum are common examples of brown algae.

RED ALGAE OR RHODOPHYCEAE

- Red algae are red because of a pigment called **Phycoerythrin.**
- Most red algae are found in **marine habitats.**
- They store food in **Floridean starch.**
- Common examples are Gracilaria, Porphyra etc.

BLUE GREEN ALGAE OR CYANOPHYCEAE

- Blue green algae are most primitive algae and are prokaryotic.
- Modern classification puts them in Kingdom Monera along with bacteria.

ECONOMIC IMPORTANCE OF ALGAE

- There are many species of **blue-green algae** capable of **fixing atmospheric nitrogen** inthe soil and are **used as biofertilizers.**
- Common examples are **Anabaena and Nostoc.**
- **Anabaena, in association with water fern Azolla** contributes nitrogen and also enrichessoils with organic matter.
- Many green algae such as **Chlorella, Ulva, Caulerpa, Enteromorpha**, etc. are used as food.
- **Chlorella** has about **50% protein and 20% of lipid and carbohydrates**. Chlorella alsoyields an **antibiotic chlorellin.**
- **Agar** is obtained from the **Red algae Gracilaria** and Agar is **used as a culture medium** forgrowing of microbes in labs. Agar is also used in **Food and Pharmaceuticals**.
- **Carragineen** which is used in the Dairy industry is obtained from a **red alga calledChondrus crispus.** It is also **used in cosmetics and Pharma.**
- **Alginic acid,** which is used as a **stabilizer and thickening agent** is obtained **fromLaminaria, the brown algae.**
- **Dynamite** is prepared with the **cell walls of Diatoms.**
- Brown algae Laminaria is a good source of Iodine.

- **Macrocystis algae are source of Potash. It's a brown algae (phaeophyceae) and is largest** algae among all.
- Some algae are extremely poisonous to fishes.
- The blue-green alga **Microcystis secretes hydroxylamine which kills aquatic life** while
- **Lyngbya and Chlorella may cause skin allergies** in human beings.
- The red alga **Cephaleuros virescens** causes **Red Rust of Tea**.
- **Algal blooms**: where excess of nutrients are available to them and floats on the watersurface and look like foam or soap lather. It is called water bloom.
    - Examples: Microcystis, Anabaena, Oscillatoria, etc.
- **Color of Red Sea: Blue green alga Trichodesmium** grows profusely is called Red Sea. It isdue to the **presence of red Phycoerythrin** in the cells of Trichodesmium.

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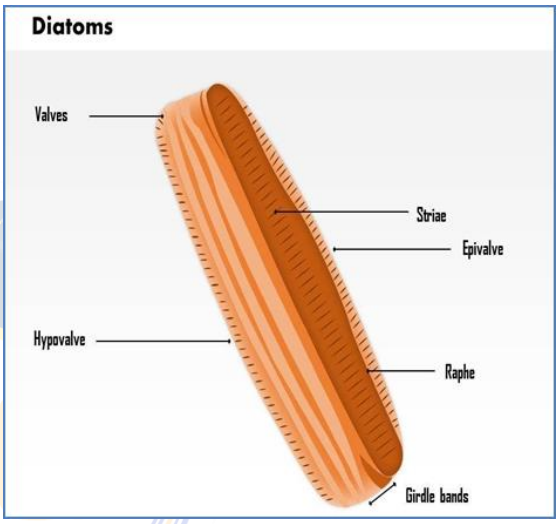
MICROORGANISM

CONTENT

- Chrysophytes
- Dinoflagellates
- Slime Moulds
- Eutrophication
- Fungi
- Virus
- Prions
- Viroids
- Vrion

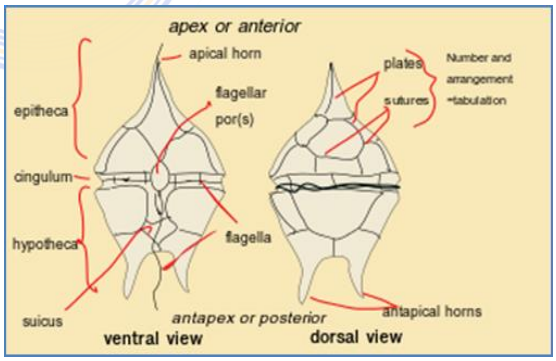
CHRYSOPHYTES

- This group includes **diatoms** and **golden algae (desmids)**.
- Most of them are **photosynthetic**.
- **Diatoms** are the chief 'producers' in the oceans.
- They are found in fresh wateras well as in marine environments.
- They are microscopic andfloat passively in water currents (**plankton**).
- In diatoms the cell walls form two thin overlapping shells.
- The walls are embedded withsilica and thus the walls are indestructible.
- Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as **'diatomaceousearth'**.



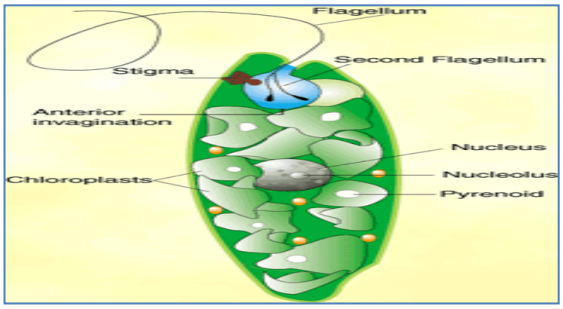
DINOFLAGELLATES

- Very often, **red dinoflagellates** (Example: Gonyaulax) undergo such rapid multiplication that they make the **sea appear red (red tides)**.



EUGLENOIDS

- Instead of a cell wall, they have a protein rich layer called **pellicle** which makes their body flexible.
- Though they are **photosynthetic** in the presence of sunlight, when deprived of sunlightthey behave like **heterotrophs** by predating on other smaller organisms.



SLIME MOULDS

- Slime moulds are **saprophytic protists**.
- Under suitable conditions, they form an aggregation called **plasmodium** which maygrow and spread over several feet.



- During unfavorable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips. The spores possess **true walls**.
- They are extremely resistant and survive for many years, even under adverse conditions. The spores are dispersed by aircurrents.

EUTROPHICATION



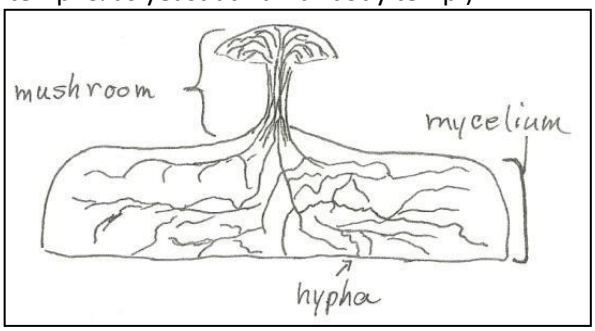
- Nitrates and phosphates are nutrients thatplants need to grow.
- In excessive amounts, however, nutrientscause a **type** of pollution called **Eutrophication**.
- **Eutrophication** stimulates an explosive growth of **algae** (**algal** blooms) that depletes the water of oxygen when the the**algae** die and are eaten by bacteria.

FUNGI

- They are eukaryotic organisms, do not contain chlorophyll. It is non-photosynthetic heterotrophs. It releases digestive enzymes to break down organic material or theirhost. They store energy as glycogen.
- Most are **saprobes/saprophytic** - live on other dead organisms.
- Most are **multicellular**, but some unicellular like **yeast**.
- Those that depend on living plants and animals are called **parasites**.
- Some fungal species live in permanent mutually dependent relationships with **bluegreenalgae (or cyanobacteria)**. Such relationships are called **symbiotic**. These symbiobic life forms are called **lichens**. They can also live as **symbionts** in association with roots of higher plants as **mycorrhiza**.
    - Fungi + Bluegreen algae (Cyanobacteria) =Lichens.
- They are **non-motile, lacks true roots, stems and leaves**. The cell wall is made of **chitin** (a complex polysaccharide).
- Lichen is the indicator of air pollution mainly sulphur dioxide.
- If air is badly polluted with sulphur dioxide there may be no lichens present, just greenalgae may be found.
- Yeasts, molds, mushrooms are the example of fungi.
- **Yeast belongs** to the division of **Ascomycetes** of Kingdom Fungi. (**Ascomycetes** are **'spore shooters'.** They are fungi **which produce microscopic spores inside special, elongated cells or sacs,** known as 'asci',)
- Most Yeasts reproduce asexually by mitosis and many do so by the asymmetricdivision process known as budding.
- Mold or mould grows in the form of **multicellular filament called hyphae.**
- Molds cause biodegradation of natural materials, which can be unwanted when itbecomes **food spoilage or damage to furniture etc**.
- **Mushroom** belongs to **Basidiomycetes** of kindom fungi. (In **basidiomycetes** the **spores develop on projections** that grow out from microscopic cells called **basidia**, rather thanbeing enveloped within cells. **In most cases the basidia are elongated and club-like)**
- They are fleshy, spore bearing fruiting body of fungus, typically produced above groundon soil or on its food source.They are extensively used in cooking.
- Amanita phalloides is also called death mushroom i.e. toxic
- **Dimorphic fungi** are fungi that exist in **form of mold and yeast** (e.g. **Penicillium marneffei,** a

human pathogen that grows as a mold at room temp. & as yeast at humanbody temp.)

- A **mycorrhiza** is a symbiotic association between a **fungus and a plant.**
- **Lichen** is a composite organism that arises from algae or cyanobacteria living filament of fungi species in mutualistic relationship.
- The body of fungus made of tiny filaments or tubes called **hyphae**.
- Hyphal networks are called mycelium.
  - Example: mushroom (Agaricus)



- The classification did not differentiate between the heterotrophic group – fungi, and the autotrophic green plants, though they also showed a characteristic difference in their walls composition – the fungi had chitin in their walls while the greenplants had a cellulosic cell wall.
- When such characteristics were considered, the fungi were placed in a separate kingdom – Kingdom Fungi. All prokaryotic organisms were grouped together underKingdom Monera and the unicellular eukaryotic organisms were placed in KingdomProtista.
- **When your bread develops a mould or your orange rots** it is because of fungi.
- The common **mushroom** you eat and toadstools are also fungi.
- White spots seen on mustard leaves are due to a parasitic fungus.
- Some unicellular fungi e.g: **yeast** is used to make bread and beer.
- In Yeast there is a division of fungus called ascomycetes also known as spore shooter.
- These ascomycetes are produced asexually & some are produced by asymmetricdivision/budding.
- Other fungi cause diseases in plants and animals; **wheat rust-causing Puccinia** is animportant example.
- Some are the source of antibiotics, e.g: **Penicillium**.
- **Penicillin**- discovered by Alexander Fleming (1st antibiotic from Penicillium Notatum)
- **Aspergillus** helps in the formation of cheese. (paneer)
- **Ergot** obtained from **Claviceps Purpurea**. It grows on Rye and other plants.
  - It causes psychological problems.
- LSD (Lysergic Acid Diethylamide)
  - Obtained from Claviceps.
  - It causes hallucinations.
- On the basis of nourishment the fungi are of three types –
- **Saprophytes:** The fungi which obtain their food or do nutrition from decayed moistleaves, moist dead wood or by some other useless rotten residues or organic substances.
- The fungi like Rhizopus, Penicillium etc are saprophytes.
- **Parasites:** The fungi which obtain their food by taking or sharing the food of any otherorganisms.
- The fungi like **Ustilago, Puccinia** etc that are harmful parasites.
- **Symbiotic:** The fungi, which coexist with other plants and facilitate water and mineralsalt and plants prepare food for them.
- The microbe lichen is the best example of symobiotic fungus.

BENEFITS OF FUNGI

- **Soil Formation and Fertility** : It decompose moist residues of leaves, dead wood, animalalong dung and other rotten organic substances into another, which act like manures, and thus soil becomes more fertile.
- **Food: Agaricus and Morchella** *are used in the forms of* **vegetables (mushrooms**) fungi.
- ***Aspergillus, penicillium are used in cheese industry, yeast a (a type of*** **fungi) likeSaccharomyces cervisiae is used in making double roti (bread dough).**
- Wines, beers are also prepared by the alcoholic fermentations of the yeasts.

- **Nitrogen fixation :** like ***Rodoturela***
- **Medicines**: antibiotics like chloromycetin, neomycin, streptomycin, teramycin *etc.*
- **Chemical Industry: Aspergillus gallomyces and Pencillium glaucum** are used in theGallic acid.
- Similarly **Gluconic acid and Fumeric acid** are prepared by the **fungi Aspergillus nigerand Rhizopus nigricans** respectively.
- Enzymes and Vitamins: The enzymes **amylase** is prepared from **Aspergillus orizae.**
- Similarly, invertase is prepared by yeasts.
- **Various vitamins like vitamin B is prepared from Streptomyces griseus.**
- **Mycoremediation : Bioremediation by means of Fungi is called Mycoremediation**. *Fungi have been shown to* **biomineralize uranium oxides**, suggesting they may have application in the bioremediation of radioactively polluted sites.
- Some fungi are **hyperaccumulators,** capable of absorbing and concentrating heavymetals in the mushroom fruit bodies.

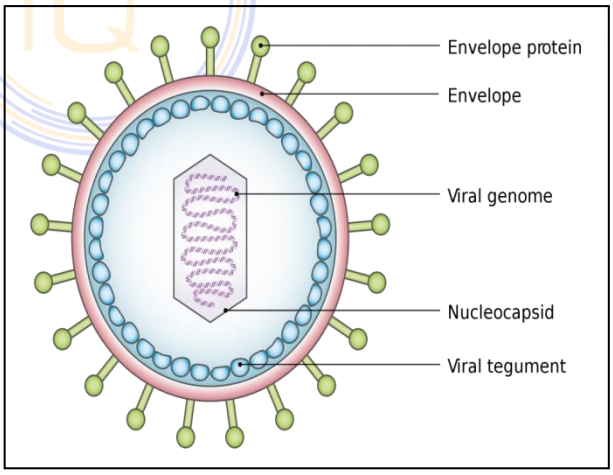
PEST CONTROL

- Beauveria bassiana, Metarhizium spp, Hirsutella spp, Paecilomyces (Isaria) spp, andLecanicillium lecanii have been used in *Pest Control.* and the yeasts, Saccaromyces cerevisiae and Schizosaccharomyces pombe, have a long history of use

COMMON FUNGAL DISEASES

- Wart disease of potato, Late blight of potato, Green ear disease of bajra, Rust of wheat, Loose smut of wheat, Tikka disease of groundnut, Red rot of sugarcane, Brown leaf spotof rice, Ergot disease of rye, Powdery mildew of wheat etc.
- Common animal and human fungal diseases include Athlete's foot scabies, Scabies, Ringworm, Meningitis, Asthma, Baldness, Aspergillosis etc.

VIRUS



- It can survive only inside another living organism, animal or plant. It is a connecting link between living and non-living things.
- Virus has genetic material, either deoxyribonucleic acid (DNA) OR ribonucleic acid (RNA). Unlike living cells, viruses can have either DNA orRNA, but not both.
- In virus, a protein coat called a **capsid** surrounds the nucleic acid. This coat serves to protect the nucleic acid and aid in its transmission between host cells.
- Some viruses have a third structure that surrounds the capsid. This is called the **envelope** and is composed of a **bilipid layer** and **glycoproteins.**
- Envelope protects the virus from immune system when host secretes anti- virussubstances called **interferons**.
- Virus was discovered by Ivanovsky.
- Tobacco Mosaic Virus was the first virus discovered.
- Capsid- protein shell of virus

LIVING PROPERTIES OF VIRUSES

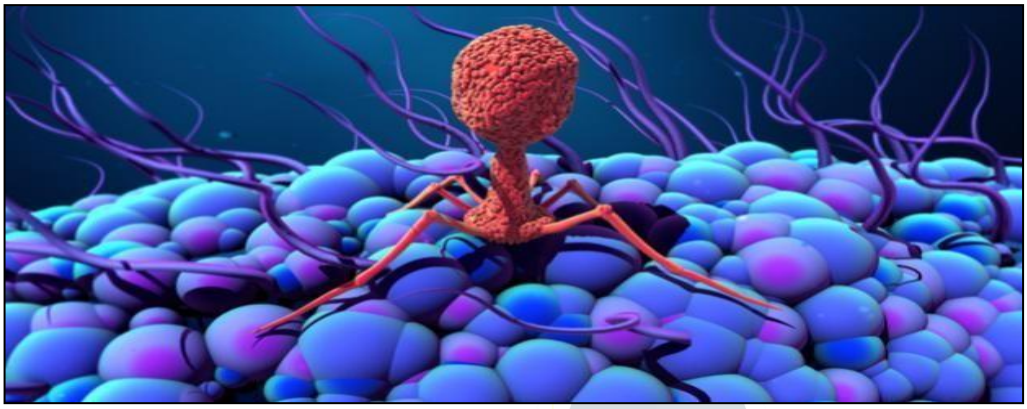
- The presence of DNA or RNA (but never both)
- Structural diversity

- Genetic and parasitic properties
- Capable of spreading the disease

NON-LIVING PROPERTIES

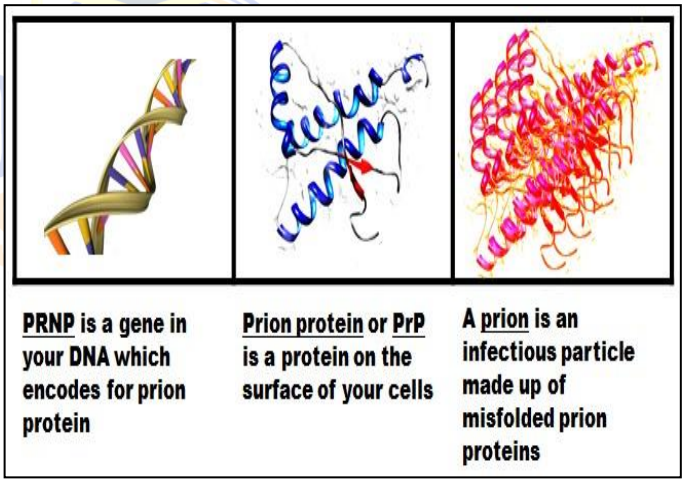
- The absence of cell and lack of protoplasm
- No any reproduction and growth outside the living cell
- Stored in the form of crystal outside the living cell
- The lack of metabolic activities like nutrition, digestion

BACTERIOPHAGE



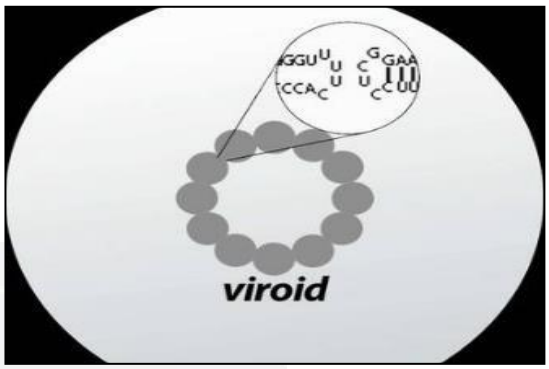
- Virus which infect the bacteria.

PRIONS



- In modern medicine certain infectious neurological diseases were found to be transmitted byan agent consisting of abnormally folded protein.
- The agent was similar in size to viruses. These agents were called prions.
- The most notable diseases caused by prions are bovine spongiform encephalopathy (BSE) commonly called mad cow disease in cattle and its analogous variant Cr– Jacobdisease (CJD) in humans.
- Prions, lack DNA, RNA the nucleic acid found in viruses.

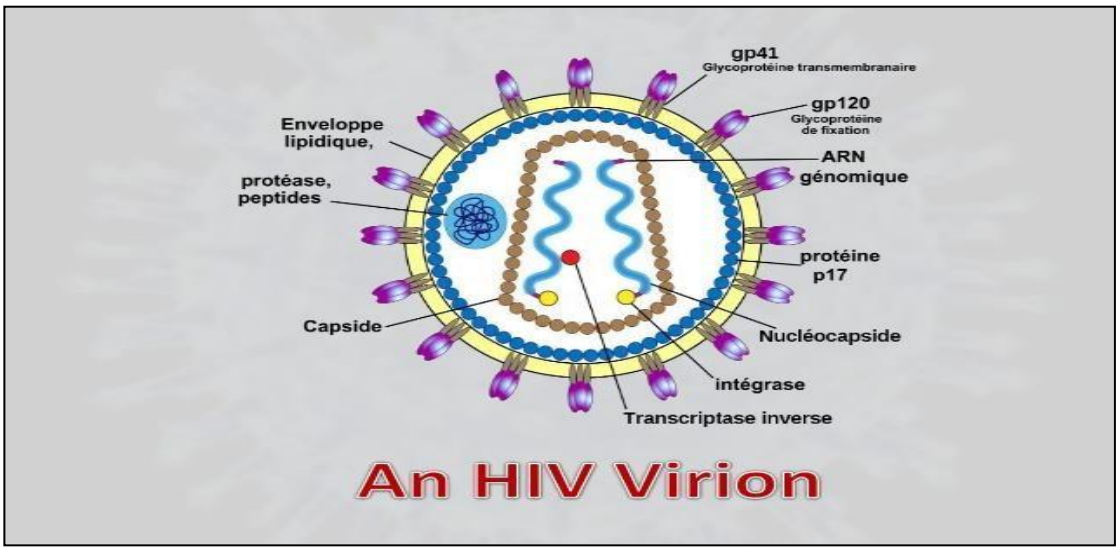
VIROIDS



- In 1971, T.O. Diener discovered anew infectious agent mainly a plant pathogens that was smaller than viruses and caused potato spindle tuber disease.
- It was found to be a free RNA;
- It lacked the protein coat that is found in viruses, hence the nameviroid.
- The RNA of the viroid was of low molecular weight.
- Extracellular, which spread the disease. But virus can be found inside or outside the cell.

VRION

Extracellular, which spread the disease. But virus can be found inside or outside the cell.



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DISEASES

CONTENT

- Introduction-Disease
- Types Of Diseases
- Diseases Spread Through Blood Transfusion
- Bacteria Diseases
- Tuberculosis ,Syphilis ,Gonnorhea ,Diphtheria
- Plague / Black Death ,Pneumonia .Typhoid
- Leprosy, Anthrax
- Bacterial Diseases Of Humans

DISEASE

- Disease is a situation where a person is not at ease, Therefore, Disease.

TYPES OF DISEASES

- On the basis of its nature, character and causes of its spreading is of two types:
  - Congenital Disease
  - Acquired Disease

CONGENITAL DISEASE

- ***Congenital Disease*** is the diseases which are present since birth.
- These are caused due to genetic abnormality or due to metabolic disorders ormalfunctioning of any organ.
- They are basically permanent, generally not easily curable and may be inherited to thechildren like Harelip, Cleft Palate, Club foot etc.
- Also due to imbalance in the chromosomes the appearance of Mongalism, are alsosome examples of it.

ACQUIRED DISEASE

- ***Acquired Disease*** is those defects or disorders which are not present by birth but appeardue to the various causes and factors. These may be further categorized into following heads:
  - ***Communicable or infectious diseases****:* These are caused by a variety of pathogenic viruses, bacteria, protozoa, fungi and worms. The pathogens aregenerally carried with the help of a vector.
  - ***Non-communicable or non-infectious or degenerative diseases*****:** These occur due to the malfunctioning of some organ or organ system in the body.
- It may be of various types like *Deficiency diseases***:** These occur due to the deficiency ofsome nutrients, minerals or vitamins, *Cancerous diseases*, *Allergy, Genetic diseases.*
- *Diseases Spread through Blood Transfusion*
- **AIDS** (Acquired Immuno Deficiency Syndrome): It destroys *immune system* of the body& is caused by the *Human Immuno deficiency Virus* (HIV).
- There are two types of HIV namely HIV-1 and HIV-2**.**
- The most common virus currently associated with AIDS is HIV-1.
- A virus found in the blood of wild African green monkey called the *Simian Immunodeficiency Virus* (SIV) is similar to HIV-2.
- HIV is a *retrovirus.* It can synthesize DNA from RNA.
- The major cell infected by HIV is the helper *T-lymphocyte* that bears the CD-4 receptorsite.

DISEASES SPREAD THROUGH BLOOD TRANSFUSION

- HIV progressively destroys *T-lymphocytes.* The patient occasionally will suffer from swollen lymph nodes, mild prolonged fever, diarrhoea or other non-specific symptoms.

IMPORTANT FACTS ABOUT AIDS

- *AIDS in India was first reported in 1986 and is the end stage of the disease.*
- *HIV antibodies can be detected by the ELISA test (Enzyme-Linked Immuno SorbentAssay).*
- *World AIDS Day is celebrated on 1st December.*

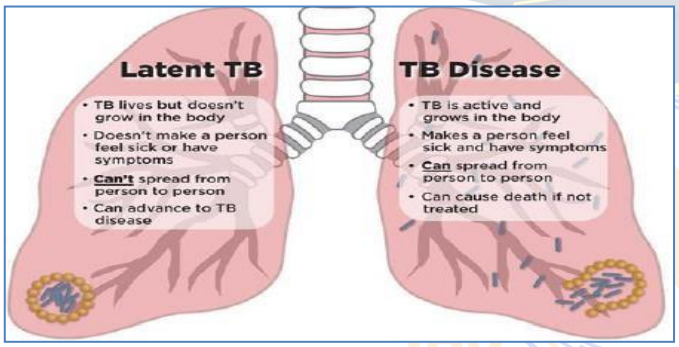
OTHERS SOURCES

- Diseases can be spread through bacteria, viruses and fungi without as well as withvectors.

BACTERIA DISEASES

- Microscopic 0.2 mm or less
- Present everywhere
- Economic importance of bacteria: Products in our day to day life are the result ofbacteria.
  - Curd, cheese
- **Antibiotic:** Streptomycin, Amoxicillin

TUBERCULOSIS



- **Tuberculosis:** By **Mycobacterium tuberculosis**

SYPHILIS

- Opurtunitic diseases
- A sexually transmitted diseases: Tryponema pallidum



GONNORHEA

- A sexually transmitted diseases

- Neisseria gonnorhoea



DIPHTHERIA

- By Corynebacterium diphtheria



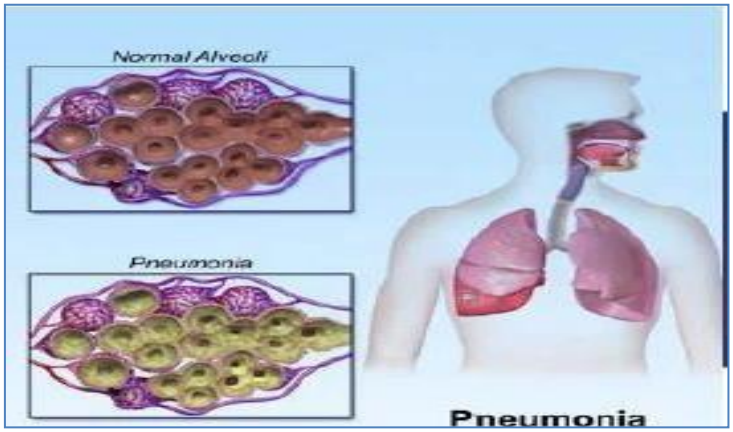
PLAGUE / BLACK DEATH

- by Yersinia Pestis
- Called black death, spread by mouse



PNEUMONIA

- by Diplococcus Pneumonia



TYPHOID

- By Salmonella typhi
- Check by Widal test
- **Bacillary dysentery** – Shigella
    - **Symptoms**: Loose motion, vomiting, nausea
- **Tetanus**- Clostridium tetani, **lock jaw** (present in Anaerobic condition), the toxin release is tetanospasmin.
- **Whooping cough** (Bordetella pertussis)

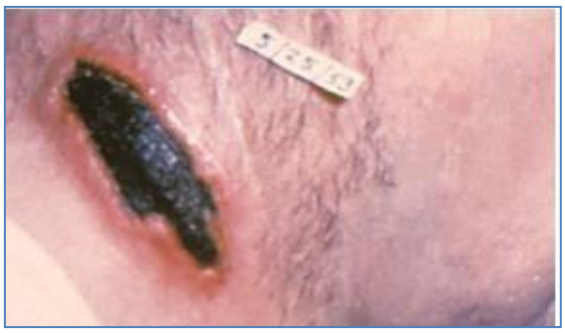
LEPROSY

- Mycobacterium leprae
- No coordination of brain,
- World leprosy day is celebrated on 30th January.



ANTHRAX

- By Bacillus Anthracis



- **Food poisoning (botulism):** Clostridium botulinum- anaerobic condition , present inpacked food like canned products.

BACTERIAL DISEASES OF HUMANS

Name of disease	Caused by	Parts of body affected	Method of spread	Type of vaccination or antibodies
Diphtheria	Corynebacterium diphtheria	Upper respiratory tract, mainly throat also toxin affects heart.	Droplet infection	Toxoid
Tuberculosis (TB)	Mycobacterium tuberculosis	Mainly lungs	Droplet infection, Drinking milk from infected cattle.	BCG living attenuated bacteria. Antibiotics e.g. streptomycin.
Whooping cough (Pertussis)	Bordetella pertussis	Upper respiratory tract, inducing violent coughing	Droplet infection	Killed bacteria

Name of disease	Caused by	Parts of body affected	Method of spread	Type of vaccination or antibodies
Gonorrhoea	Neisseria gonorrhoeae	Reproductive organs: mainly mucous membranes of urinogenital tract. Newborn infants may acquire serious eye infections if they pass through infected birth canal.	Contagion by sexual contact	Antibiotics, e.g. penicillin, streptomycin
Syphilis	Treponema pallidum	Reproductive organs, then eyes, bones, joints, central nervous system, heart and skin.	Contagion by sexual contact	Antibiotics. e.g. penicillin

Name of disease	Caused by	Parts of body affected	Method of spread	Type of vaccination or antibodies
Tetanus	Clostridium tetani	Blood. Toxin produced which affects motor nerves of spinal cord and hence muscles, causing lockjaw and spreading to the muscles.	Wound infection	Toxoid
Cholera	Vibrio cholera	Alimentary canal: mainly small intestine.	Faecal contamination (a) food - or water borne of material contaminated with faeces from infected person. (b) handling of contaminated Objects. (c) vector, e,g. flies moving from human faeces to food.	Killed bacteria: short-lived protection and not always effective Antibiotics e.g. tetracyclines, chloramphenicol.

Name of disease	Caused by	Parts of body affected	Method of spread	Type of vaccination or antibodies
Typhoid fever	Salmonella typhi	Alimentary canal, then spreading to lymph and blood, lungs, bone marrow, spleen.	Same as cholera	Killed bacteria (TAB vaccine)
Bacterial dysentery	Shigella dysenteriae	Alimentary canal, mainly ileum and colon	Same as cholera	No vaccine.
Bacterial food poisoning (gastro enteritis or salmonellosis)	Salmonella spp.	Alimentary canal	Mainly foodborne meat from infected animals from poultry and pigs. Also via faecal contamination as cholera	Antibiotic. e.g. tetracyclines.

DISEASES CAUSED BY WORMS

Disease	Pathogen	Mode of transmission	Main symptoms
Ancylostomiasis or 'Hook-worm disease'	Ancylostoma duobenale, small intestine (jejunum) of man	Transmission from person to person, filariform larvae passed out in faeces, man picks up infection walking barefoot on faecally-contaminated soil.	Dermatitis; reddish, severe anaemia; duodenal ulcer, constipation. Patient pale, face puffy with swelling of lower eyelids.
Ascariasis	Ascaris Lumbricoides; small intestine (jejunum) of man	Transmission from person to person, ripe eggs passed out in faeces, infection affected by swallowing ripe Ascaris eggs with raw vegetables.	Larvae in lung cause pneumonia. May give rise to typhoid-like fever, causes protein and Vitamin A deficiencies resulting in protein-calorie malnutrition and night blindness respectively. Can cause appendicitis, jaundice.
Disease	**Pathogen**	**Mode of transmission**	**Main symptoms**
Enterobiasis or 'Pinworm disease'	Enterobius vermicularis, caecum and vermiform appendix	Transmission from one person to another by ingestion of eggs in contaminated food or drink.	Eczematous condition round the anus, bed wetting at night, inflammation of vermiform appendix.
Filariasis	Wuchereria Bancrofti, lymphatic vessels and lymph nodes.	Part of the life cycle in mosquito in which larvae develop and become infectious to man, with mosquito bite larvae deposited on skin which enter through puncture wound and reach lymphatic channels	Elephantiasis i.e. enormous enlargement of certain parts such as that of leg, scrotum, penis, labia, clitoris, breast, forearm.

DISEASES IN PLANTS

- **Citrus canker**- Xanthomonas citri



- **Bacterial wilt of potato or ring diseases** – Pseudomonas solanacearum



BACTERIAL BLIGHT OF RICE (XANTHOMONAS ORYZAE)

- It causes wilting of seedlings and yellowing and drying of leaves.



- **Black arm of cotton**- Xanthomonas campestris



PATHOGEN

- Corynobacterium tritici (the disease isknown to occur onlyin association with nematode (Anguina tritici))



SYMPTOMS

- The disease is characterized by twisting, distortion ofear-head and rotting of spinkelet's, with profuse oozing of yellow liquid from the infected tissues.
- **Bacterial canker of tomato**: Clavibacter michiganense

VIRUS

- **Polio virus**- Poliomyelitis
- **Herpes simplex**- Herpes simplexvirus
- Causes diseases only in plants andanimals
- **Chicken pox**: Varicella virus
- **Small pox**: Variola virus (extinctfrom world)
- **Common cold**: Rhino virus
- **Shingles/ Herpes zoster** is an infections caused by the Varicella- zoster virus, which is the same virusthat causes chicken pox.
- **Measles**: Caused by infection with the Rubeola virus, a paramyxovirus.
- **Mumps**- Mumps virus (swollenparotid gland)







- **Dengue** – Arbovirus by aedes mosquito, the platelets count decreases.
- Symptoms of dengue fever include severe joint and muscle pain, swollen lymphnodes,headache, fever, exhaustion, and rash.

DISEASES IN PLANTS

- **TOBACCO MOSAIC VIRUS**
- Chlorophyll of the leaves gets destroyed
- **Bunchy top of banana –** the banana bunchy top virus (BBTV)
- **Leaf Curl Of Papaya**



FUNGI

- Non green plants
- Grow on dead and decaying organic matter and absorb food from it
- Microscopic form to large ones with big fruiting bodies
- Common example is mushroom.
- Poisonous mushroom - **Toadstool.**

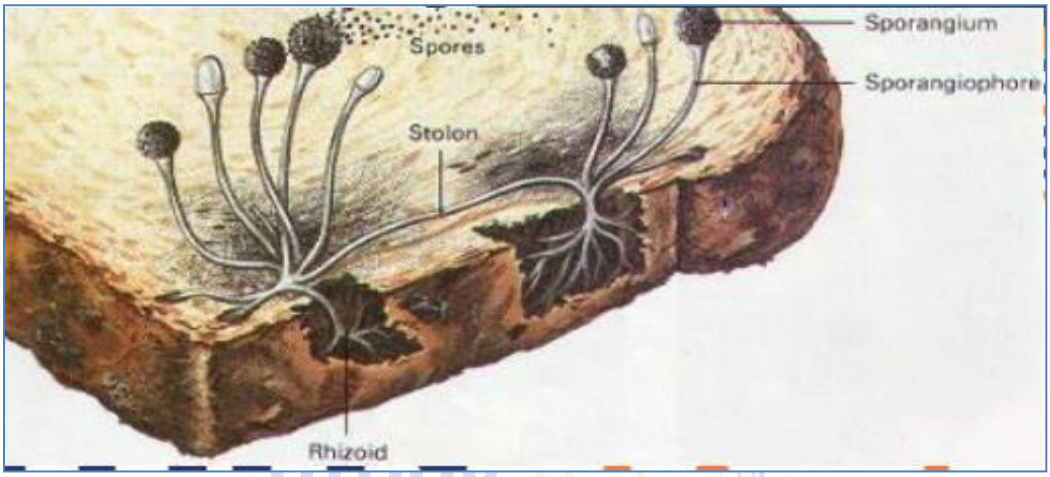




Disease	Pathogen responsible	Mode of transmission	Main symptoms
Ringworm (tinea)	Microsporum, Trichophyton	Direct contact from unbathed cats and dogs or objects handled by infected individuals	Contain one or more blistered areas on skin and scalp. Cause partial and temporary baldness in children.
Athlete's foot	Trichophyton	Bad foot hygiene where skin remains warm and moist for long periods.	Painful itching or burning sensation in the infected areas. Crack appears in the skin, mass of loose dead skin clings between toes.

Disease	Pathogen responsible	Mode of transmission	Main symptoms
Madura foot	Maurella Mycetomi	Fungi gain entry through some minor injury to the skin.	Produce a chronic, granulating infection of the lower extremities, affected part becomes enlarged and develops many deep sores, extensive bone destruction leading to crippling deformities.
Dhobie itch	Several Different Fungi	Direct contact through Objects handled by infected person.	A type of ringworm infection usually located in the groin and inner surfaces of thighs, red rash which itches intensely.

- **Bread Mould** – Aspergillus, Penicillium, Mucor etc**.**
- On the leaves of Sarso, Muli etc black rounded spots.



USEFUL

- Mushroom, soy sauce, tofu, tempeh, miso
- Ganoderma lucidum- a fungus used for preventing cancer,tea powder, toothpaste
- Yeast- Baking, Brewing(bread, beer, wine)
- Antibiotics – Penicillin and Cephalosporin
- Cheese – Roquefort and Camembert

HARMFUL

- Crop diseases- eg: Potato blight (cause of Irish famine in 1845)- Phytophthora infestan
- Dandruff in human being: Malassezia furfur
- White rust of cucifers: Albugo candida
- Loose smut of wheat: Ustilago tritici
- Tikka disease of ground nut are caused by Cercospora personata and Cercosporaarachidicola.
- The wart disease of potato/ black scab are caused by Synchytrium endobioticum.
- The brown leaf spot or sesame leaf spot: Bipolaris oryzae
- Atheletes foot- Trichophytan sp. Tinea pedis e.g.: Ringworm – Trichophytan sp.
- Damping off of seedlings
- Red rot of sugarcane – Colletotrichum Falcatum

VECTOR BORNE DISEASES

- **Kala-azar**: Kala-azar is a slow progressing indigenous diseases caused by a protozoanparasite of genus leishmania.

- **Leishmania donovani** is the only parasite causing this disease in india. Recurrent fever:skin-dry, thin and scaly and may be lost.
- Persons show grayish discoloration of the skin of hands, feet, abdomen and face whichgives the Indian name kala azar meaning **BLACK FEVER.**

DENGUE

- Dengue is a viral disease. It is transmitted by the infective bite of aedes aegyptimosquito.

MALARIA

- It is a mosquito-borne infectious diseases of humans caused by Eukaryotic protists ofthe genus plasmodium. It is spreaded by female anopheles mosquitoes.
- **ETIOLOGY**
  - Malaria is caused by obligate intracellular protozoan paasites of the genus
  - "plasmodium". The four species of human malarial parasites are: plasmodium
  - vivax
  - Plasmodium falciparm
  - Plasmodium malariae (black water fever)
  - Plasmodium ovale

SYMPTOMS OF MALARIA

- Skin: Chills, Sweating
- Respiratory: Dry cough
- Spleen: Enlargement
- Stomach: Nausea, vomiting
- Central: Headache
- Systemic: Fever
- Muscular: Fatigue, Pain
- Back: Pain
- Chikungunya

WHAT IS CHIKUNGUNYA?

- Manifest with crippling arthritic disease of sudden onset.
- Caused by Virus, genus: Alpha virus
- Chikungunya viral infection manifests with febrile illness.
- Aedes aegypti mosquito
- Mosquitoes bites infect the humans

VIRAL DISEASES OF HUMANS

Name of disease	Caused by	Parts of body affected	Method of spread	Type of Vaccination
Influenza	A myxovirus (RNA virus)	Respiratory passages: epithelial lining of trachea and bronchi.	Droplet Infection	Killed virus
Common cold	Large variety of viruses, commonly rhino-virus (RNA Virus)	Respiratory passages	Droplet Infection	Intramuscular injection.
Smallpox	Variola virus (DNA virus)	Respiratory passages, then skin	Droplet Infection (Wounds in skin)	Living atteneuated virus applied by scratching skin, no longer carried

Name of disease	Caused by	Parts of body affected	Method of spread	Type of Vaccination
Poliomyelitis (polio)	Poliovirus (RNA Virus)	Pharynx and intestines, then blood; occasionally motor neurons in spinal cord, paralysis may occur.	Droplet infection or via human faeces	Living attenuated virus given orally
Yellow fever	An arbovirus i.e arthropod-borne virus (RNA Virus)	Lining of blood vessels and liver	Vector- arthropods e.g ticks, mosquitoes	Living attenuated virus
AIDS	Retrovirus (RNA virus)	Skin Cancer	Sexual intercourse homo- and hetrosexuals	Not available

Name of disease	Caused by	Parts of body affected	Method of spread	Type of Vaccination
Ebola haemorrhagic fever	Ebola Virus disease (EVD)	Fatal Illness in Humans, Fever	It is transmitted to people from wild animals and spreads in the human population through human-to-human transmission.	No licensed Ebola vaccine is available
Zika disease	Zika Virus (mosquito borne disease)	Causes mild illness in the people like dengue, yellow fever	Basically Infection in pregnant women is linked to abnormally small heads in their babies.	No vaccine available

OTHER DISEASES

- **Cancer:** They are characterized by the uncontrolled growth and division of cells which leads to a mass of cells known as neoplasm. Abnormal and persistent cell division localized in a particular region is called benign tumor.
- **Gout:** It results from accumulation of *uric acid* crystals in the *synovial joints.* It is adisease associated with an inborn error of uric acid metabolism that increases production or interferes with the excretion of uric acid.

Disease	*Deficiency*	*Symptoms*
Anaemia	Haemoglobin (iron)	General weakness and pale complexion
Goitre	Iodine	Painful joints
Beri-beri	Vitamin B_1 (Thiamine)	Weakness, swelling and pain in legs, loss of appetite, enlarged heart
Scurvy	Vitamin C (Ascorbic acid)	Swollen gums, delayed wound healing
Rickets	Vitamin D	Sleeplessness, pale face, diarrhoea, deformed skull, pelvis and limbs in children
Hypokalemia	Potassium	Rise in heartbeat, kidney damage, weakness
Night blindness	Vitamin A	–
Xerophthalmia	Vitamin A	Dryness
Dermatosis	Vitamin A	Skin diseases
Ariboflavinosis	Vitamin B_2 (Riboflavin)	Blurred vision, soreness of eyes and tongue
Pellagra	Nicotinic Acid (Vitamin B complex)	Diarrhoea, mental lethargy, red skin, itchy hands, feet, elbows and knees

- **Haemophilia** is called bleeder's disease. It is a disorder which is sex-linked and is arecessive condition. In a patient of haemophilia, *blood clotting* is deficient.
- **Haemophilia A,** is characterized by lack of anti- haemophilia globulin factor VIII. Aboutfour-fifths of the cases of haemophilia are of this type.
- **Haemophilia B** or Christmas disease results from a defect in plasma thromboplasticcomponent.
- **Hepatitis:** It is a viral disease, causes hepatic anorexia resulting in liver damage (liver cancer) or jaundice. It is transmitted by the faecal-oral route. Children and young adultsare susceptible to it and no vaccines are acceptable.

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